



5.1 Rates of change and the derivative

Calculus · IB Diploma · Concept guide and original exam-style practice

Understand the concept

The derivative $f'(x)$ gives the instantaneous rate of change of f , or equivalently the gradient of the tangent to $y=f(x)$ at that point; it can be defined from first principles as $f'(x)=\lim_{h \rightarrow 0} [f(x+h)-f(x)]/h$. Standard rules include the power rule $d/dx(x^n)=nx^{n-1}$, and derivatives of e^{kx} , $\ln x$, $\sin x$ and $\cos x$, along with the chain, product and quotient rules for combined functions. Derivatives are used to find gradients of tangents/normals and instantaneous rates of change in applied contexts.

Worked example: follow the reasoning

Find the equation of the tangent to $y=x^3-2x+1$ at $x=2$.

Solution: $y'=3x^2-2$; at $x=2$, gradient $=3(4)-2=10$. $y(2)=8-4+1=5$. Tangent: $y-5=10(x-2) \rightarrow y=10x-15$.

Quick check · multiple choice

Find dy/dx for $y=5x^4$.

- A $5x^3$
- B $20x^3$
- C $4x^3$
- D $20x^4$

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 The height of a projectile above the ground, $h(t)$ metres, t seconds after launch, is [6]
 $h(t)=-4.9t^2+30t+2$. (a) Find $h'(t)$, the vertical velocity function. (b) Find the velocity at $t=2$ s and state whether the projectile is rising or falling at this time. (c) Find the time at which the projectile reaches its maximum height.

Final answer: _____



Answers and explanation

Question 1 [6]

(a) $h'(t) = -9.8t + 30$. (b) $h'(2) = -19.6 + 30 = 10.4$ m/s; since this is positive, the projectile is rising at $t = 2$ s. (c) Maximum height occurs when $h'(t) = 0$: $-9.8t + 30 = 0 \rightarrow t = 30/9.8 \approx 3.06$ s.

Quick check answer

B. Using the power rule, $dy/dx = 4 \times 5x^3 = 20x^3$.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Mathematics AI subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.